Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA (Remote)

PROFESSIONAL SUMMARY

AI engineer who **built AgentProbe end-to-end** - the exact system Hightouch's AI Agents team ships: **production data agents** with multi-step reasoning (LangGraph Orchestrator, RAG Tracer, Auto Fix), **production LLM pipelines** across multiple providers (Anthropic, OpenAI, Groq, Ollama), and a **5-metric evaluation engine** with CI/CD regression gates. Built from prototype to open-source - not a demo. **8+ years production Python**; full-stack delivery across 120+ Agile sprints at Entrata (100K+ users, 99.5% uptime, 1.6M records); AI on FHIR demonstrates building AI for domain-specific enterprise workflows. The exact "rapidly prototype, then architect to production" pattern Hightouch describes. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Production Data Agents (LangGraph) | LLM Pipeline Architecture | Multi-Provider LLM Routing | Prototype → Production (open-source) | Python (8+ years production) | RAG Pipelines & Vector Search (ChromaDB) | LLM Evaluation Framework (LLM-as-Judge) | FastAPI / REST Platform Layer | AWS / Docker / CI/CD | TypeScript / JavaScript / React | Enterprise-Scale Backend (1M+ records, 100K+ MAU) | AI Architecture Decisions

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated data with careful output validation and quality monitoring.
- Built and delivered **20+ full-stack modules** (Python, TypeScript/JavaScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users** - startup pace with enterprise-grade uptime (99.5% over 6+ months).
- Led Docker-based containerization for 30%+ of microservices, cutting execution time by **50%**; improved production query performance by **30%** across 200+ SQL queries; **1M+ property records** at consistent scale.
- Migrated client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents** across 100K+ users; managed EC2, RDS, and Glue.
- Built **5+ internal platform tools** adopted across squads, reducing resolution time by **40%**; drove secure coding standards and peer code review practices across all releases.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students; designed exercises, drove iterative quality improvement - the same customer-first thinking Hightouch's role requires when translating user workflows into AI-assisted experiences.

PROJECTS

AgentProbe: Production Data Agents Platform - LLM Pipelines, Multi-Agent, Evaluation OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built the same system Hightouch's AI Agents team ships - production-grade LLM pipelines and data agents, from prototype to open-source. **Production data agents**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) using multi-step reasoning and tool calls - agents that operate on data, answer questions, and self-correct without human intervention. **LLM pipeline architecture**: multi-provider routing across Anthropic, OpenAI, Groq, Ollama - dynamic selection by task type, latency, and capability, with per-run regression tracking. **Prototype to production**: greenfield → open-source with 15-endpoint FastAPI backend + React observability dashboard - the exact "rapidly prototype, then architect to production" pattern Hightouch describes. **Evaluation**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD gates - agents must be reliable and auditable from day one. **RAG pipeline**: ChromaDB + embeddings + retrieval quality scoring - agents use company data to answer complex questions across multi-step runs.
github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: Enterprise Domain-Specific Agents - LLMs + FHIR APIs

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

AI agents for clinical decision support - domain-specific agents answering natural language queries against FHIR APIs via LLMs, BERT, and embeddings. Same pattern as Hightouch's data agents answering analytics questions using company data. **HIPAA-compliant RBAC. ~60% retrieval time reduction.** Built for a regulated enterprise domain. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, LLM APIs (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates, Prompt Engineering, Context Engineering, Token Management

Backend & Infra: FastAPI, Flask, React, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum, Claude Code, GitHub Copilot